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Operating System

13Week [02 December - 08 December]

SOC3010 OS QUIZ6 FALL 2024

Started on	2024-12-13 18:28
State	Finished
Completed on	2024-12-13 23:51
Time taken	5 hours 23 mins
Marks	1243.00/1253.00
Grade	29.76 out of 30.00 (99%)

Question 1

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"At any time, only one pending signal of a given type may exist for a process; additional pending signals of the same type to the same process are not queued but simply discarded."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question 2

Correct

Mark 5.00 out of 5.00

The stores general information about a specific file. For disk-based filesystems, this object usually corresponds to a file control block stored on disk.

Select one:

- ☐ 1. dentry object
- ☒ 2. inode object ✓
- ☐ 3. superblock object
- ☐ 4. file object

The correct answer is: inode object

Question 3

Correct

Mark 5.00 out of 5.00

In journaling mode, only changes to filesystem metadata are logged to the journal and is the fastest mode.

Answer: writeback ✓

The correct answer is: writeback

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Question 4

Correct

Mark 5.00 out of 5.00

..... field in file object provides Pointer to file operation table

Select one:

- ☐ 1. f_dentry
- ☒ 2. f_op ✓
- ☐ 3. f_list
- ☐ 4. f_vfsmnt

The correct answer is: f_op

Question 5

Correct

Mark 5.00 out of 5.00

The field of the memory region object or descriptor provides First linear address after that memory region.

Select one:

- ☐ 1. vm_next
- ☐ 2. vm_mm
- ☒ 3. vm_end ✓
- ☐ 4. vm_start

The correct answer is: vm_end

Question 6

Correct

Mark 5.00 out of 5.00

Given a memory area MA with start address 0x70000000 and size 364 MB, compute the end address

Select one:

- ☐ 1. 0x83FFFFFF
- ☐ 2. 0x867FFFFF
- ☒ 3. 0x86BFFFFF ✓
- ☐ 4. 0x85FFFFFF

The correct answer is: 0x86BFFFFF

Question 7

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"Every red-black tree with n internal nodes has a height of at most $2 \times \log(n + 1)$."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **8**

Correct

Mark 5.00 out of 5.00

The function is invoked to destroy a slab and release the corresponding page frames to the zoned page frame allocator.

Select one:

- ☐ 1. kmem_freepages()
- ☐ 2. kmem_getpages()
- ☒ 3. slab_destroy() ✓
- ☐ 4. cache_grow()

The correct answer is: slab_destroy()

Question **9**

Correct

Mark 5.00 out of 5.00

State whether the following statements are TRUE or FALSE.

“Several processes may have read locks on some file region, but only one process can have a write lock on it at the same time.” ..

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **10**

Correct

Mark 5.00 out of 5.00

..... state of dentry object indicates that the inode associated with the dentry does not exist because the corresponding disk inode has been deleted and the d_inode field of the dentry object is set to NULL.

Answer: ✓

The correct answer is: Negative

Question **11**

Correct

Mark 5.00 out of 5.00

..... field in superblock object provides Pointers for a list of superblock objects of a given filesystem type

Select one:

- ☐ 1. s_type
- ☒ 2. s_instances ✓
- ☐ 3. s_fs_info
- ☐ 4. s_id

The correct answer is: s_instances

Question **12**

Correct

Mark 5.00 out of 5.00

If of 3-bit error_code parameter of do_page_fault() function is clear, the exception was caused by an access to a page that is not present (the Present flag in the Page Table entry is clear).

Select one:

- ☒ 1. bit 0 ✓
- ☐ 2. bit 2
- ☐ 3. bit 1
- ☐ 4. bit 3

The correct answer is: bit 0

Question **13**

Correct

Mark 5.00 out of 5.00

Kernel uses function to establish a permanent kernel mapping.

Select one:

- ☐ 1. kunmap()
- ☒ 2. kmap() ✓
- ☐ 3. kunmap_atomic
- ☐ 4. kmap_atomic()

The correct answer is: kmap()

Question **14**

Correct

Mark 5.00 out of 5.00

..... field in inode object provides pointer to superblock object

Answer: ✓

The correct answer is: i_sb

Question **15**

Correct

Mark 5.00 out of 5.00

..... system call changes the heap size of the process.

Select one:

- ☐ 1. mmap_cache
- ☐ 2. munmap()
- ☒ 3. brk() ✓
- ☐ 4. mmap_cache

The correct answer is: brk()

Question **16**

Correct

Mark 5.00 out of 5.00

The directory on which a filesystem is mounted is called the

Answer: ✓

The correct answer is: mount point.

Question **17**

Correct

Mark 5.00 out of 5.00

In Linux, the function used for Finding a free interval is

Select one:

- ☐ 1. find_vma()
- ☒ 2. get_unmapped_area() ✓
- ☐ 3. insert_vm_struct()
- ☐ 4. find_vma_intersection()

The correct answer is: get_unmapped_area()

Question **18**

Correct

Mark 5.00 out of 5.00

Categorize the Coda file system into class.

Select one:

- ☐ 1. Special file system
- ☐ 2. None of these classes
- ☐ 3. Disk-based file system
- ☒ 4. Network file system ✓

The correct answer is: Network file system

Question **19**

Correct

Mark 5.00 out of 5.00

..... field in dentry object provides pointer to Inode associated with filename.

Select one:

- ☐ 1. d_op
- ☐ 2. d_lru
- ☐ 3. d_sb
- ☒ 4. d_inode ✓

The correct answer is: d_inode

Question **20**

Correct

Mark 5.00 out of 5.00

Categorize the NCP file system into class.

Select one:

- ☐ 1. Disk-based file system
- ☒ 2. Network file system ✓
- ☐ 3. Special file system
- ☐ 4. None of these classes

The correct answer is: Network file system

Question **21**

Correct

Mark 5.00 out of 5.00

Categorize the JFS file system into class.

Select one:

- ☐ 1. Network file system
- ☒ 2. Disk-based file system ✓
- ☐ 3. Special file system
- ☐ 4. None of these classes

The correct answer is: Disk-based file system

Question **22**

Correct

Mark 5.00 out of 5.00

The dentry hash table is implemented by means of a dentry_hashtable array in Linux. The array's size usually depends on the amount of RAM installed in the system; the default value is entries per megabyte of RAM.

Select one:

- ☐ 1. 64
- ☐ 2. 512
- ☒ 3. 256 ✓
- ☐ 4. 128

The correct answer is: 256

Question **23**

Correct

Mark 5.00 out of 5.00

Categorize the AFS file system into class.

Select one:

- ☐ 1. Disk-based file system
- ☒ 2. Network file system ✓
- ☐ 3. Special file system
- ☐ 4. None of these classes

The correct answer is: Network file system

Question **24**

Correct

Mark 5.00 out of 5.00

By default, a process may own up to different memory regions.

Select one:

- ☐ 1. 4096
- ☐ 2. 32768
- ☒ 3. 65536 ✓
- ☐ 4. 65535

The correct answer is: 65536

Question **25**

Correct

Mark 5.00 out of 5.00

The field of the memory descriptor provides maximum number of page frames ever owned by the process.

Select one:

- ☐ 1. mmlist
- ☒ 2. hiwater_rss ✓
- ☐ 3. total_vm
- ☐ 4. rss

The correct answer is: hiwater_rss

Question **26**

Correct

Mark 5.00 out of 5.00

Kernel uses function to establish a temporary kernel mapping.

Select one:

- ☐ 1. kunmap()
- ☒ 2. kmap_atomic() ✓
- ☐ 3. kunmap_atomic()
- ☐ 4. kmap()

The correct answer is: kmap_atomic()

Question **27**

Correct

Mark 5.00 out of 5.00

Given a main memory of 8GB, determine the size of the mem_map array required to store all the page descriptors in memory.

Select one:

- ☐ 1. 64KB
- ☐ 2. 64GB
- ☐ 3. 128MB
- ☒ 4. 64MB ✓

The correct answer is: 64MB

Question **28**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"If the page does not have any access rights, the Present bit is cleared so that each access generates a Page Fault exception. "

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **29**

Correct

Mark 5.00 out of 5.00

..... operation associated with file object sends a command to an underlying hardware device. This method applies only to device files.

Select one:

- ☐ 1. poll(file, poll_table)
- ☒ 2. ioctl(inode, file, cmd, arg) ✓
- ☐ 3. fsync(file, dentry, flag)
- ☐ 4. mmap(file, vma)

The correct answer is: ioctl(inode, file, cmd, arg)

Question **30**

Correct

Mark 5.00 out of 5.00

Because processes are created and destroyed quite frequently, the kernel wastes time allocating and deallocating the page frames containing the same memory areas repeatedly; the allows these to be saved in a cache and reused quickly thereby improving the performance.

Select one:

- ☐ 1. Buddy System
- ☐ 2. Zone Allocator
- ☐ 3. Per-CPU Page frame cache
- ☒ 4. Slab Allocator ✓

The correct answer is: Slab Allocator

Question **31**

Correct

Mark 5.00 out of 5.00

State whether the following statements are TRUE or FALSE.

"It is possible to get a write lock when another process owns a read lock for the same file region, and vice versa"

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **32**

Correct

Mark 5.00 out of 5.00

In the reverse operation, page frames assigned to a slab can be released by invoking the function.

Select one:

- ☐ 1. slab_destroy()
- ☐ 2. kmem_getpages()
- ☒ 3. kmem_freepages() ✓
- ☐ 4. cache_grow()

The correct answer is: kmem_freepages()

Question **33**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"The problem of non-allocation of high-memory page frames does not exist on 64-bit hardware platforms, because the available linear address space is much larger than the amount of RAM that can be installed in short, the ZONE_HIGHMEM zone of these architectures is always empty."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **34**

Correct

Mark 5.00 out of 5.00

..... system call can be used to lock an arbitrary region of a file (even a single byte) or to lock the whole file (including data appended in the future).

Answer: ✓

The correct answer is: fcntl()

Question **35**

Correct

Mark 5.00 out of 5.00

Kernel uses function to destroy a temporary kernel mapping

Select one:

- ☐ 1. kmap()
- ☒ 2. kunmap_atomic() ✓
- ☐ 3. kunmap()
- ☐ 4. kmap_atomic()

The correct answer is: kunmap_atomic()

Question **36**

Correct

Mark 5.00 out of 5.00

New Slab objects may be allocated by invoking the function.

Select one:

- ☐ 1. kmem_cache_free()
- ☐ 2. vmalloc()
- ☐ 3. vfree()
- ☒ 4. kmem_cache_alloc() ✓

The correct answer is: kmem_cache_alloc()

Question **37**

Correct

Mark 5.00 out of 5.00

Each filesystem consists of two kinds of blocks: those containing the so-called and those containing regular data.

Answer: ✓

The correct answer is: metadata

Question **38**

Correct

Mark 5.00 out of 5.00

Categorize the HPFS file system into class.

Select one:

- ☐ 1. None of these classes
- ☒ 2. Disk-based file system ✓
- ☐ 3. Network file system
- ☐ 4. Special file system

The correct answer is: Disk-based file system

Question **39**

Correct

Mark 5.00 out of 5.00

In Linux, the function used for Inserting a region in the memory descriptor list is

Select one:

- ☐ 1. find_vma()
- ☒ 2. insert_vm_struct() ✓
- ☐ 3. get_unmapped_area()
- ☐ 4. find_vma_intersection()

The correct answer is: insert_vm_struct()

Question **40**

Correct

Mark 5.00 out of 5.00

Maximum number of memory nodes in NUMA 32-bit architecture is

Select one:

- ☒ 1. 64 ✓
- ☐ 2. 32
- ☐ 3. 256
- ☐ 4. 1024

The correct answer is: 64

Question **41**

Correct

Mark 5.00 out of 5.00

In Linux, the function used for Finding a region that overlaps a given interval is

Select one:

- ☐ 1. get_unmapped_area()
- ☐ 2. find_vma()
- ☐ 3. insert_vm_struct()
- ☒ 4. find_vma_intersection() ✓

The correct answer is: find_vma_intersection()

Question **42**

Correct

Mark 5.00 out of 5.00

..... operation associated with file object Checks whether there is activity on a file and goes to sleep until something happens on it.

Select one:

- ☐ 1. fsync(file, dentry, flag)
- ☐ 2. ioctl(inode, file, cmd, arg)
- ☒ 3. poll(file, poll_table) ✓
- ☐ 4. mmap(file, vma)

The correct answer is: poll(file, poll_table)

Question **43**

Correct

Mark 5.00 out of 5.00

" Mark the files as candidates for mandatory locking by setting their set-group bit (SGID) and clearing the group-execute permission bit." SAY TRUE OR FALSE.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **44**

Correct

Mark 5.00 out of 5.00

..... macro is used to request 2^{order} contiguous page frames. It returns the address of the descriptor of the first allocated page frame or returns NULL if the allocation failed.

Select one:

- ☐ 1. `__get_free_page(gfp_mask)`
- ☐ 2. `alloc_page(gfp_mask)`
- ☐ 3. `__get_free_pages(gfp_mask, order)`
- ☒ 4. `alloc_pages(gfp_mask, order)` ✓

The correct answer is: `alloc_pages(gfp_mask, order)`

Question **45**

Correct

Mark 5.00 out of 5.00

To get better system performance, is used to quickly satisfy the allocation requests for single page frames.

Select one:

- ☐ 1. Slab Allocator
- ☐ 2. Buddy System
- ☐ 3. Zone Allocator
- ☒ 4. Per-CPU Page frame cache ✓

The correct answer is: Per-CPU Page frame cache

Question **46**

Correct

Mark 5.00 out of 5.00

The children of a red node in RB tree must be

Select one:

- ☐ 1. yellow
- ☒ 2. black ✓
- ☐ 3. blue
- ☐ 4. red

The correct answer is: black

Question **47**

Correct

Mark 5.00 out of 5.00

----- system call creates a memory mapping for a file, thus enlarging the process address space.

Select one:

- ☐ 1. `brk()`
- ☒ 2. `mmap()` ✓
- ☐ 3. `mmap_cache`
- ☐ 4. `munmap()`

The correct answer is: `mmap()`

Question **48**

Correct

Mark 5.00 out of 5.00

The function destroys all the slabs in a cache by invoking `slab_destroy()` iteratively

Select one:

- ☐ 1. `kmem_cache_init()`
- ☐ 2. `kmem_freepages()`
- ☐ 3. `kmem_cache_create()`
- ☒ 4. `kmem_cache_shrink()` ✓

The correct answer is: `kmem_cache_shrink()`

Question **49**

Correct

Mark 5.00 out of 5.00

Ctrl & BACKSLASH key combinations generate signal.

Select one:

- ☒ 1. `SIGQUIT` ✓
- ☐ 2. `SIGKILL`
- ☐ 3. `SIGINT`
- ☐ 4. `SIGSTP`

The correct answer is: `SIGQUIT`

Question **50**

Correct

Mark 5.00 out of 5.00

The function modifies the size of the heap directly; the `addr` parameter specifies the new value of `current->mm->brk`, and the return value is the new ending address of the memory region (the process must check whether it coincides with the requested `addr` value).

Select one:

- ☐ 1. `sys_brk(addr)`
- ☒ 2. `brk(addr)` ✓
- ☐ 3. `realloc(ptr,size)`
- ☐ 4. `calloc(n,size)`

The correct answer is: `brk(addr)`

Question **51**

Correct

Mark 5.00 out of 5.00

Ctrl & Z key combinations generate signal.

Select one:

- ☒ 1. `SIGSTP` ✓
- ☐ 2. `SIGKILL`
- ☐ 3. `SIGINT`
- ☐ 4. `SIGQUIT`

The correct answer is: `SIGSTP`

Question **52**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"Like a semaphore used to protect a critical region in code, the lock is considered "advisory" because it doesn't work unless other processes cooperate in checking the existence of a lock before accessing the file. "

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **53**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"vm_end - vm_start denotes the length of the memory region."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **54**

Correct

Mark 5.00 out of 5.00

Linux stores memory descriptors in data structures called for efficient insertion, deletion & search operations.

Select one:

- ☐ 1. Singly linked list
- ☐ 2. Hash Table
- ☐ 3. Doubly linked list
- ☒ 4. Red-black trees ✓

The correct answer is: Red-black trees

Question **55**

Correct

Mark 5.00 out of 5.00

..... operation is used as second argument of flock system call to remove lock on the file.

Select one:

- ☐ 1. LOCK_EX
- ☐ 2. SET_LCK
- ☒ 3. LOCK_UN ✓
- ☐ 4. LOCK_SH

The correct answer is: LOCK_UN

Question **56**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"Each memory region descriptor identifies a linear address interval."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **57**

Correct

Mark 5.00 out of 5.00

Categorize the NFS file system into class.

Select one:

- ☐ 1. Special file system
- ☐ 2. Disk-based file system
- ☒ 3. Network file system ✓
- ☐ 4. None of these classes

The correct answer is: Network file system

Question **58**

Correct

Mark 5.00 out of 5.00

Mount the filesystem where mandatory locking is required using the option in the command.

Select one:

- ☐ 1. -o writeback, umount
- ☐ 2. -o writeback, mount
- ☒ 3. -o mand, mount ✓
- ☐ 4. -o mand, umount

The correct answer is: -o mand, mount

Question **59**

Correct

Mark 5.00 out of 5.00

..... operation associated with inode object creates a new hard link that refers to the file specified by old_dentry in the directory dir; the new hard link has the name specified by new_dentry.

Select one:

- ☐ 1. follow_link(inode, nameidata)
- ☒ 2. link(old_dentry, dir, new_dentry) ✓
- ☐ 3. mknod(dir, dentry, mode, rdev)
- ☐ 4. lookup(dir, dentry, nameidata)

The correct answer is: link(old_dentry, dir, new_dentry)

Question **60**

Correct

Mark 10.00 out of 10.00

A process cannot use more than file descriptors and kernel enforces a dynamic bound on the maximum number of file descriptors in a process to

Select one:

- ☐ 1. 1048576, 512
- ☒ 2. 1048576, 1024 ✓
- ☐ 3. 65536, 1024
- ☐ 4. 65536, 512

The correct answer is: 1048576, 1024

Question **61**

Correct

Mark 5.00 out of 5.00

..... field in superblock object provides pointer to Filesystem type structure

Select one:

- ☒ 1. s_type ✓
- ☐ 2. s_op
- ☐ 3. s_id
- ☐ 4. s_fs_info

The correct answer is: s_type

Question **62**

Correct

Mark 5.00 out of 5.00

The field in the process descriptor points to the memory descriptor owned by the process.

Select one:

- ☐ 1. active_mm
- ☒ 2. mm ✓
- ☐ 3. mmdrop()
- ☐ 4. mmput()

The correct answer is: mm

Question **63**

Correct

Mark 5.00 out of 5.00

Maximum number of memory nodes in NUMA 64-bit architecture is

Select one:

- ☐ 1. 32
- ☒ 2. 1024 ✓
- ☐ 3. 256
- ☐ 4. 64

The correct answer is: 1024

Question **64**

Correct

Mark 5.00 out of 5.00

In 80x86 architecture, in the fourth gigabyte of RAM, PKMAP_BASE refers to address

Select one:

- ☐ 1. 0x10000000
- ☐ 2. 0xC0000000
- ☐ 3. 0xFC000000
- ☒ 4. 0xFE000000 ✓

The correct answer is: 0xFE000000

Question **65**

Correct

Mark 5.00 out of 5.00

The area of main memory that contains a cache is divided into each of which consists of one or more contiguous page frames that contain both allocated and free objects.

Answer: ✓

The correct answer is: slabs

Question **66**

Incorrect

Mark 0.00 out of 10.00

On UMA 64-bit architectures, flags field in the page descriptor has bits for the zone index and bits for the node number.

Select one:

- ☐ 1. 2,4
- ☐ 2. 2, 1
- ☒ 3. 2, 10 ✗
- ☐ 4. 2, 6

The correct answer is: 2, 1

Question **67**

Correct

Mark 3.00 out of 3.00

..... system call allows a process to lock a whole file and not a region of the file.

Answer: ✓

The correct answer is: flock()

Question **68**

Correct

Mark 5.00 out of 5.00

For each process, the kernel maintains a variable that points to the top of the heap.

Select one:

- ☐ 1. pc
- ☐ 2. cs
- ☐ 3. sp
- ☒ 4. brk ✓

The correct answer is: brk

Question **69**

Correct

Mark 5.00 out of 5.00

The function decreases the mm_users field of a memory descriptor and invokes mmdrop() function to release the memory descriptor.

Select one:

- ☐ 1. mmdrop()
- ☐ 2. active_mm
- ☐ 3. mm
- ☒ 4. mmpu() ✓

The correct answer is: mmpu()

Question **70**

Correct

Mark 5.00 out of 5.00

Categorize the Pipe file system (pipefs) into class.

Select one:

- ☐ 1. Disk-based file system
- ☐ 2. Network file system
- ☒ 3. Special file system ✓
- ☐ 4. None of these classes

The correct answer is: Special file system

Question **71**

Correct

Mark 5.00 out of 5.00

..... operation is used as second argument of flock system call to apply read lock on the file.

Select one:

- ☒ 1. LOCK_SH ✓
- ☐ 2. SET_LCK
- ☐ 3. LOCK_UN
- ☐ 4. LOCK_EX

The correct answer is: LOCK_SH

Question **72**

Correct

Mark 5.00 out of 5.00

In 32-bit architecture, the zone includes page frames that cannot be directly accessed by the kernel through the linear mapping in the fourth gigabyte of linear address space.

Select one:

- ☒ 1. ZONE_HIGHMEM ✓
- ☐ 2. ZONE_DMA
- ☐ 3. ZONE_NORMAL
- ☐ 4. ZONE_LOWMEM

The correct answer is: ZONE_HIGHMEM

Question **73**

Correct

Mark 5.00 out of 5.00

..... operation associated with inode object creates a new disk inode for a special file associated with a dentry object in some directory. The mode and rdev parameters specify, respectively, the file type and the device's major and minor numbers.

Select one:

- ☐ 1. lookup(dir, dentry, nameidata)
- ☐ 2. follow_link(inode, nameidata)
- ☒ 3. mknod(dir, dentry, mode, rdev) ✓
- ☐ 4. link(old_dentry, dir, new_dentry)

The correct answer is: mknod(dir, dentry, mode, rdev)

Question **74**

Correct

Mark 5.00 out of 5.00

When a process in User Mode invokes the brk() system call, the kernel executes the function.

Select one:

- ☐ 1. sbrk(incr)
- ☒ 2. sys_brk(addr) ✓
- ☐ 3. realloc(ptr,size)
- ☐ 4. calloc(n,size)

The correct answer is: sys_brk(addr)

Question **75**

Correct

Mark 5.00 out of 5.00

The macro yields the address of the page descriptor associated with the page frame having number pfn.

Select one:

- ☐ 1. virt_to_page(addr)
- ☒ 2. pfn_to_page(pfn) ✓
- ☐ 3. page_to_virt(pfn)
- ☐ 4. page_to_pfn(addr)

The correct answer is: pfn_to_page(pfn)

Question **76**

Correct

Mark 5.00 out of 5.00

Specific caches are created by the function

Select one:

- ☐ 1. kmem_cache_shrink()
- ☐ 2. kmem_cache_init()
- ☐ 3. kmem_freepages()
- ☒ 4. kmem_cache_create() ✓

The correct answer is: kmem_cache_create()

Question **77**

Correct

Mark 5.00 out of 5.00

When the slab allocator creates a new slab, it relies on the zoned page frame allocator to obtain a group of free contiguous page frames by invoking the function

Select one:

- ☒ 1. kmem_getpages() ✓
- ☐ 2. kmem_freepages()
- ☐ 3. cache_grow()
- ☐ 4. slab_destroy()

The correct answer is: kmem_getpages()

Question **78**

Correct

Mark 5.00 out of 5.00

When the kernel must delete a linear address interval from the address space of the current process, it uses the function.

Select one:

- ☒ 1. do_munmap() ✓
- ☐ 2. vma_unlink()
- ☐ 3. vma_link()
- ☐ 4. do_mmap()

The correct answer is: do_munmap()

Question **79**

Correct

Mark 5.00 out of 5.00

The root of the RB tree must be

Select one:

- ☐ 1. yellow
- ☒ 2. black ✓
- ☐ 3. blue
- ☐ 4. red

The correct answer is: black

Question **80**

Correct

Mark 5.00 out of 5.00

A kernel function gets dynamic memory by invoking to use the slab allocator for specialized or general-purpose objects

Select one:

- ☐ 1. vmalloc() or vmalloc_32()
- ☐ 2. alloc_pages
- ☒ 3. kmem_cache_alloc() or kmalloc() ✓
- ☐ 4. __get_free_pages()

The correct answer is: kmem_cache_alloc() or kmalloc()

Question **81**

Correct

Mark 5.00 out of 5.00

..... field in dentry object provides pointer to Superblock object of the file.

Select one:

- ☐ 1. d_lru
- ☐ 2. d_op
- ☐ 3. d_inode
- ☒ 4. d_sb ✓

The correct answer is: d_sb

Question **82**

Correct

Mark 5.00 out of 5.00

In Linux the system calls must be invoked by executing the Assembly instruction.

Select one:

- ☐ 1. int \$0x90
- ☐ 2. int \$0x128
- ☐ 3. int \$0x180
- ☒ 4. int \$0x80 ✓

The correct answer is: int \$0x80

Question **83**

Correct

Mark 5.00 out of 5.00

..... field in superblock object provides pointer to Superblock methods structure.

Select one:

- ☐ 1. s_fs_info
- ☐ 2. s_id
- ☐ 3. s_type
- ☒ 4. s_op ✓

The correct answer is: s_op

Question **84**

Correct

Mark 5.00 out of 5.00

The kernel subsystem that handles the memory allocation requests for groups of contiguous page frames is called the

Select one:

- ☒ 1. Zoned page frame allocator ✓
- ☐ 2. Slab Allocator
- ☐ 3. Buddy System
- ☐ 4. Per-CPU Page frame cache

The correct answer is: Zoned page frame allocator

Question **85**

Correct

Mark 5.00 out of 5.00

The `mm_alloc()` calls function in turn to store and initialize the new memory descriptor in a slab allocator cache.

Select one:

- ☐ 1. `mm_count`
- ☒ 2. `kmem_cache_alloc()` ✓
- ☐ 3. `mm_users`
- ☐ 4. `mm_alloc()`

The correct answer is: `kmem_cache_alloc()`

Question **86**

Correct

Mark 5.00 out of 5.00

..... system call destroys a memory mapping for a file, thus contracting the process address space.

Select one:

- ☐ 1. `mmap()`
- ☒ 2. `munmap()` ✓
- ☐ 3. `brk()`
- ☐ 4. `mmap_cache`

The correct answer is: `munmap()`

Question **87**

Correct

Mark 5.00 out of 5.00

A kernel function gets dynamic memory by invoking to get a noncontiguous memory area.

Select one:

- ☐ 1. `__get_free_pages()`
- ☐ 2. `kmalloc()`
- ☒ 3. `vmalloc()` or `vmalloc_32()` ✓
- ☐ 4. `kmem_cache_alloc()`

The correct answer is: `vmalloc()` or `vmalloc_32()`

Question **88**

Correct

Mark 5.00 out of 5.00

The `ext2_truncate()` function essentially scans the disk inode's `i_block` array to locate all data blocks and all blocks used for the indirect addressing. These blocks are then released by repeatedly invoking

Select one:

- ☐ 1. `ext2_get_block()`
- ☐ 2. `ext2_alloc_block()`
- ☐ 3. `ext2_truncate()`
- ☒ 4. `ext2_free_blocks()` ✓

The correct answer is: `ext2_free_blocks()`

Question **89**

Correct

Mark 5.00 out of 5.00

Categorize the `/dev` into class.

Select one:

- ☐ 1. Disk-based file system
- ☒ 2. Not a file system ✓
- ☐ 3. Special file system
- ☐ 4. Network file system

The correct answer is: Not a file system

Question **90**

Correct

Mark 5.00 out of 5.00

Given a memory area `MA1` with start address `0xC0000000` and size **256 MB**, compute the end address

Select one:

- ☐ 1. `0xEFFFFFFF`
- ☐ 2. `0xFFFFFFFF`
- ☒ 3. `0xCFFFFFFF` ✓
- ☐ 4. `0xBFFFFFFF`

The correct answer is: `0xCFFFFFFF`

Question **91**

Correct

Mark 5.00 out of 5.00

The function removes a memory region from the linked list and the red-black tree of the memory descriptor.

Select one:

- ☒ 1. `vma_unlink()` ✓
- ☐ 2. `split_vma()`
- ☐ 3. `do_munmap()`
- ☐ 4. `vma_link()`

The correct answer is: `vma_unlink()`

Question **92**

Correct

Mark 10.00 out of 10.00

If the amount of physical memory that is directly mapped in the kernel's fourth gigabyte of linear addresses is 1024MB, then the reserved pool size used for atomic memory allocation requests to be used only on low-on memory conditions is

Select one:

- ☐ 1. 65536KB
- ☐ 2. 32684KB
- ☐ 3. 1024KB
- ☒ 4. 128KB ✓

The correct answer is: 128KB

Question **93**

Correct

Mark 5.00 out of 5.00

Categorize the NTFS file system into class.

Select one:

- ☐ 1. Special file system
- ☐ 2. None of these classes
- ☐ 3. Network file system
- ☒ 4. Disk-based file system ✓

The correct answer is: Disk-based file system

Question **94**

Correct

Mark 5.00 out of 5.00

The field of the memory descriptor provides size of the process address space (number of pages).

Select one:

- ☒ 1. total_vm ✓
- ☐ 2. rss
- ☐ 3. mmlist
- ☐ 4. hiwater_rss

The correct answer is: total_vm

Question **95**

Correct

Mark 5.00 out of 5.00

..... field in the zone descriptor Specifies how many page frames in each zone must be reserved for handling low-on-memory critical situations.

Select one:

- ☐ 1. pages_high
- ☐ 2. pages_low
- ☐ 3. pages_min
- ☒ 4. lowmem_reserve ✓

The correct answer is: lowmem_reserve

Question **96**

Correct

Mark 5.00 out of 5.00

The function differs from the other heap management functions because it is the only one implemented as a system call.

Select one:

- ☒ 1. brk() ✓
- ☐ 2. calloc(n,size)
- ☐ 3. sys_brk(addr)
- ☐ 4. realloc(ptr,size)

The correct answer is: brk()

Question **97**

Correct

Mark 5.00 out of 5.00

Frequent requests and releases of groups of contiguous page frames of different sizes may lead to a situation in which several small blocks of free page frames are "scattered" inside blocks of allocated page frames. As a result, it may become impossible to allocate a large block of contiguous page frames, even if there are enough free pages to satisfy the request. this is a well-known memory management problem called

Select one:

- ☒ 1. external fragmentation ✓
- ☐ 2. internal fragmentation
- ☐ 3. contiguous memory allocation
- ☐ 4. non-contiguous memory allocation

The correct answer is: external fragmentation

Question **98**

Correct

Mark 5.00 out of 5.00

The stores information about the linking of a directory entry (that is, a particular name of the file) with the corresponding file. Each disk-based filesystem stores this information in its own particular way on disk.

Select one:

- ☒ 1. dentry object ✓
- ☐ 2. inode object
- ☐ 3. file object
- ☐ 4. superblock object

The correct answer is: dentry object

Question **99**

Correct

Mark 5.00 out of 5.00

The function requests an array consisting of n elements of size size; if the allocation succeeds, it initializes the array components to 0 and returns the linear address of the first element.

Select one:

- ☐ 1. sys_brk(addr)
- ☐ 2. realloc(ptr,size)
- ☒ 3. calloc(n,size) ✓
- ☐ 4. brk()

The correct answer is: calloc(n,size)

Question **100**

Correct

Mark 5.00 out of 5.00

The of a process consists of all linear addresses that the process is allowed to use.

Answer: ✓

The correct answer is: address space

Question **101**

Correct

Mark 5.00 out of 5.00

The field of the memory descriptor provides pointers to adjacent elements in the list of memory descriptors

Select one:

- ☐ 1. hiwater_rss
- ☐ 2. rss
- ☒ 3. mmlist ✓
- ☐ 4. total_vm

The correct answer is: mmlist

Question **102**

Correct

Mark 5.00 out of 5.00

..... is used to store all information related to the process address space.

Select one:

- ☒ 1. Memory Descriptor ✓
- ☐ 2. Page Descriptor
- ☐ 3. Process Descriptor
- ☐ 4. File descriptor or inode

The correct answer is: Memory Descriptor

Question **103**

Correct

Mark 5.00 out of 5.00

The function implements the buddy system strategy for freeing page frames.

Select one:

- ☐ 1. __rmqueue()
- ☐ 2. buffered_rmqueue()
- ☒ 3. __free_pages_bulk() ✓
- ☐ 4. free_hot_cold_page()

The correct answers are: __free_pages_bulk(), buffered_rmqueue()

Question **104**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"The pages_min field of the zone descriptor stores the number of reserved page frames inside the zone. This field plays also a role for the page frame reclaiming algorithm, together with the pages_low and pages_high fields. The pages_low field is always set to 5/4 of the value of pages_min, and pages_high is always set to 3/2 of the value of pages_min."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **105**

Correct

Mark 5.00 out of 5.00

Given a main memory of 4GB, determine the size of the mem_map array required to store all the page descriptors in memory.

Select one:

- ☐ 1. 32KB
- ☐ 2. 32GB
- ☐ 3. 4MB
- ☒ 4. 32MB ✓

The correct answer is: 32MB

Question **106**

Correct

Mark 5.00 out of 5.00

The function is invoked during system initialization to set up the general caches.

Select one:

- ☐ 1. kmem_cache_shrink()
- ☐ 2. kmem_freepages()
- ☒ 3. kmem_cache_init() ✓
- ☐ 4. kmem_cache_create()

The correct answer is: kmem_cache_init()

Question **107**

Correct

Mark 5.00 out of 5.00

The stores information about the interaction between an open file and a process. This information exists only in kernel memory during the period when a process has the file open.

Select one:

- ☐ 1. inode object
- ☐ 2. superblock object
- ☒ 3. file object ✓
- ☐ 4. dentry object

The correct answer is: file object

Question **108**

Correct

Mark 5.00 out of 5.00

..... field in file object provides a pointer to the Mounted filesystem containing the file.

Select one:

- ☒ 1. f_vfsmnt ✓
- ☐ 2. f_list
- ☐ 3. f_dentry
- ☐ 4. f_op

The correct answer is: f_vfsmnt

Question **109**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"To store the memory regions of a process, Linux uses both a linked list and a red-black tree. Both data structures contain pointers to the same memory region descriptors."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **110**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"Objects that have the same offset within different slabs will, with a relatively high probability, end up mapped in the same cache line. The cache hardware might therefore waste memory cycles transferring two objects from the same cache line back and forth to different RAM locations, while other cache lines go underutilized."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **111**

Correct

Mark 5.00 out of 5.00

The field in the process descriptor points to the memory descriptor used by the process when it is in execution.

Select one:

- ☐ 1. mm
- ☒ 2. active_mm ✓
- ☐ 3. mmdrop()
- ☐ 4. mmput()

The correct answer is: active_mm

Question **112**

Correct

Mark 5.00 out of 5.00

Categorize the ADFS file system into class.

Select one:

- ☐ 1. None of these classes
- ☒ 2. Disk-based file system ✓
- ☐ 3. Special file system
- ☐ 4. Network file system

The correct answer is: Disk-based file system

Question **113**

Correct

Mark 5.00 out of 5.00

How many dentry objects are created by the kernel when looking up the pathname /usr/tmp/test.

Select one:

- ☒ 1. 3 ✓
- ☐ 2. 1
- ☐ 3. 4
- ☐ 4. 2

The correct answer is: 3

Question **114**

Correct

Mark 5.00 out of 5.00

The function is used in Buddy system to find a free block in a zone.

Select one:

- ☒ 1. `__rmqueue()` ✓
- ☐ 2. `__free_pages_bulk()`
- ☐ 3. `free_hot_cold_page()`
- ☐ 4. `buffered_rmqueue()`

The correct answer is: `__rmqueue()`

Question **115**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"If the page has both Write and Share access rights, the Read/Write bit is reset."

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **116**

Correct

Mark 5.00 out of 5.00

" To make use of mandatory locks, mandatory locking must be enabled both on the filesystem that contains the file to be locked, and on the file itself. "

SAY TRUE OR FALSE

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **117**

Correct

Mark 5.00 out of 5.00

The function allocates page frames through the per-CPU page frame caches

Select one:

- ☐ 1. `__rmqueue()`
- ☒ 2. `buffered_rmqueue()` ✓
- ☐ 3. `__free_pages_bulk()`
- ☐ 4. `free_hot_cold_page()`

The correct answer is: `buffered_rmqueue()`

Question **118**

Correct

Mark 5.00 out of 5.00

..... state of dentry object indicates that the dentry object is currently used by the kernel. The d_count usage counter is positive, and the d_inode field points to the associated inode object. The dentry object contains valid information and cannot be discarded.

Answer:



The correct answer is: In Use

Question **119**

Correct

Mark 5.00 out of 5.00

..... field in superblock object provides Pointer to superblock information of a specific filesystem.

Select one:

- ☐ 1. s_type
- ☐ 2. s_id
- ☐ 3. s_op
- ☒ 4. s_fs_info ✓

The correct answer is: s_fs_info

Question **120**

Correct

Mark 5.00 out of 5.00

..... field in dentry object provides pointer to Inode associated with filename.

Select one:

- ☐ 1. d_lru
- ☒ 2. d_inode ✓
- ☐ 3. d_sb
- ☐ 4. d_op

The correct answer is: d_inode

Question **121**

Correct

Mark 5.00 out of 5.00

The function decreases mm_count field of memory descriptor and, if it becomes zero, releases the mm_struct data structure, i.e., memory descriptor.

Select one:

- ☒ 1. mmdrop() ✓
- ☐ 2. mmput()
- ☐ 3. mm
- ☐ 4. active_mm

The correct answer is: mmdrop()

Question **122**

Correct

Mark 5.00 out of 5.00

A signal is not delivered as long as it is

Select one:

- ☐ 1. fatal
- ☐ 2. ignored
- ☐ 3. unblocked
- ☒ 4. blocked ✓

The correct answer is: blocked

Question **123**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"When the user types a command at the console, the shell process creates a new process to execute the command. As a result, a fresh address space, and thus a set of memory regions, is assigned to the new process."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **124**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"If the NX bit is supported and the page does not have the Execute access right, then NX bit is set."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **125**

Correct

Mark 5.00 out of 5.00

Categorize the AFFS file system into class.

Select one:

- ☒ 1. Disk-based file system ✓
- ☐ 2. Network file system
- ☐ 3. Special file system
- ☐ 4. None of these classes

The correct answer is: Disk-based file system

Question **126**

Correct

Mark 5.00 out of 5.00

Categorize the CIFS file system into class.

Select one:

- ☐ 1. Disk-based file system
- ☐ 2. None of these classes
- ☐ 3. Special file system
- ☒ 4. Network file system ✓

The correct answer is: Network file system

Question **127**

Correct

Mark 5.00 out of 5.00

The stores information concerning a mounted filesystem. For disk-based filesystems, this object usually corresponds to a filesystem control block stored on disk.

Select one:

- ☐ 1. inode object
- ☐ 2. dentry object
- ☒ 3. superblock object ✓
- ☐ 4. file object

The correct answer is: superblock object

Question **128**

Correct

Mark 5.00 out of 5.00

The function, which is the Page Fault interrupt service routine for the 80x86 architecture which compares the linear address that caused the Page Fault against the memory regions of the current process to check access right violations.

Select one:

- ☒ 1. do_page_fault() ✓
- ☐ 2. do_mmap() function
- ☐ 3. vma_unlink() function
- ☐ 4. do_munmap()

The correct answer is: do_page_fault()

Question **129**

Correct

Mark 5.00 out of 5.00

Given a memory area MA1 with start address 0xF0000000 and size 256 MB, compute the end address

Select one:

- ☐ 1. 0xEFFFFFFF
- ☐ 2. 0xBFFFFFFF
- ☐ 3. 0xCFFFFFFF
- ☒ 4. 0xFFFFFFFF ✓

The correct answer is: 0xFFFFFFFF

Question 130

Correct

Mark 5.00 out of 5.00

Categorize the /proc file system into class.

Select one:

- ☐ 1. None of these classes
- ☐ 2. Network file system
- ☒ 3. Special file system ✓
- ☐ 4. Disk-based file system

The correct answer is: Special file system

Question 131

Correct

Mark 5.00 out of 5.00

..... field in the node descriptor provides pointer to array of page descriptors of the node.

Select one:

- ☐ 1. node_zonelists
- ☒ 2. node_mem_map ✓
- ☐ 3. node_zones
- ☐ 4. node_start_pfn

The correct answer is: node_mem_map

Question 132

Correct

Mark 5.00 out of 5.00

The utility program is invoked by the boot script to check the value stored in s_mount_state field of the superblock on disk; if it is not equal to EXT2_VALID_FS, the filesystem was not properly unmounted, and therefore starts checking all disk data structures of the filesystem.

(WRITE ONE ANSWER FOR BOTH)

Answer: ✓

The correct answer is: e2fsck

Question 133

Correct

Mark 5.00 out of 5.00

..... state of dentry object indicates that the dentry object contains no valid information and is not used by the VFS. The corresponding memory area is handled by the slab allocator.

Answer: ✓

The correct answer is: Free

Question **134**

Correct

Mark 5.00 out of 5.00

..... field in inode object provides Number of hard links

Answer:



The correct answer is: i_nlink

Question **135**

Correct

Mark 5.00 out of 5.00

..... in each memory zone stores page frames for DMA operation.

Select one:

- ☐ 1. A hot cache
- ☐ 2. Unified Cache
- ☐ 3. Split Cache
- ☒ 4. A cold cache ✓

The correct answer is: A cold cache

Question **136**

Correct

Mark 5.00 out of 5.00

..... operation associated with inode object translates a symbolic link specified by an inode object; if the symbolic link is a relative pathname, the lookup operation starts from the directory specified in the second parameter.

Select one:

- ☒ 1. follow_link(inode, nameidata) ✓
- ☐ 2. mknod(dir, dentry, mode, rdev)
- ☐ 3. link(old_dentry, dir, new_dentry)
- ☐ 4. lookup(dir, dentry, nameidata)

The correct answer is: follow_link(inode, nameidata)

Question **137**

Correct

Mark 5.00 out of 5.00

..... field in file object provides dentry object associated with the file.

Select one:

- ☐ 1. f_list
- ☒ 2. f_dentry ✓
- ☐ 3. f_op
- ☐ 4. f_vfsmnt

The correct answer is: f_dentry

Question **138**

Correct

Mark 5.00 out of 5.00

Signals that have been generated but not yet delivered are called signals .

Answer: ✓

The correct answer is: pending

Question **139**

Correct

Mark 5.00 out of 5.00

A problem called is caused by a mismatch between the size of the memory request and the size of the memory area allocated to satisfy the request.

Select one:

- ☐ 1. contiguous memory allocation
- ☐ 2. non-contiguous memory allocation
- ☒ 3. internal fragmentation ✓
- ☐ 4. external fragmentation

The correct answer is: internal fragmentation

Question **140**

Correct

Mark 5.00 out of 5.00

The field of the memory descriptor provides number of memory regions in the process address space.

Select one:

- ☐ 1. mm_rb
- ☒ 2. map_count ✓
- ☐ 3. mmap
- ☐ 4. mmap_cache

The correct answer is: map_count

Question **141**

Correct

Mark 30.00 out of 30.00

Identify all the meta data in an Ext2 file system :

Select one or more:

- ☐ 1. free blocks
- ☒ 2. inode bitmap blocks ✓
- ☐ 3. data blocks
- ☒ 4. data bitmap blocks ✓
- ☒ 5. superblocks ✓
- ☒ 6. group block descriptors ✓
- ☒ 7. inodes ✓
- ☒ 8. blocks used for indirect addressing (indirection blocks) ✓

The correct answers are: superblocks, group block descriptors, inodes, blocks used for indirect addressing (indirection blocks), data bitmap blocks, inode bitmap blocks

Question **142**

Correct

Mark 5.00 out of 5.00

The size of a DMA zone is ,..... .

Select one:

- ☐ 1. 896MB
- ☐ 2. 256MB
- ☒ 3. 16MB ✓
- ☐ 4. 128MB

The correct answer is: 16MB

Question **143**

Correct

Mark 5.00 out of 5.00

A signal is for a given process if delivering the signal causes the kernel to kill the process.

Select one:

- ☐ 1. unblocked
- ☒ 2. fatal ✓
- ☐ 3. blocked
- ☐ 4. ignored

The correct answer is: fatal

Question **144**

Correct

Mark 5.00 out of 5.00

The slab allocator assigns a new slab to the cache by invoking the function

Select one:

- ☐ 1. kmem_freepages()
- ☒ 2. cache_grow() ✓
- ☐ 3. slab_destroy()
- ☐ 4. kmem_getpages()

The correct answer is: cache_grow()

Question **145**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"A process may create an IPC-shared memory region to share data with other cooperating processes. In this case, the kernel assigns a new memory region to the process to implement this construct."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **146**

Correct

Mark 5.00 out of 5.00

..... field in inode object provides pointer to inode operations

Answer: ✓

The correct answer is: i_op

Question **147**

Correct

Mark 5.00 out of 5.00

Given a memory area MA1 with start address 0x80000000 and size 1024 MB, compute the end address

Select one:

- ☐ 1. 0xEFFFFFFF
- ☐ 2. 0xCFFFFFFF
- ☒ 3. 0xBFFFFFFF ✓
- ☐ 4. 0xEFFFFFFF

The correct answer is: 0xBFFFFFFF

Question **148**

Correct

Mark 5.00 out of 5.00

To mount an Ext3 filesystem stored in the /dev/sda1 partition on the / mount point with the ordered mode, the system administrator can type the command:

mount -t ext3 -o data=..... /dev/sda2 /

Answer: ✓

The correct answer is: ordered

Question **149**

Correct

Mark 5.00 out of 5.00

The field of the memory descriptor stores the number of lightweight processes that share the mm_struct data structure.

Select one:

- ☐ 1. mm_count
- ☒ 2. mm_users ✓
- ☐ 3. kmem_cache_alloc()
- ☐ 4. mm_alloc()

The correct answer is: mm_users

Question **150**

Correct

Mark 5.00 out of 5.00

All page descriptors are stored in the array in dynamic memory.

Answer: ✓

The correct answer is: mem_map

Question **151**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"A single API function could make several system calls."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **152**

Correct

Mark 5.00 out of 5.00

..... field in dentry object provides pointer to Dentry methods.

Select one:

- ☐ 1. d_inode
- ☐ 2. d_lru
- ☒ 3. d_op ✓
- ☐ 4. d_sb

The correct answer is: d_op

Question **153**

Correct

Mark 5.00 out of 5.00

The field of the memory descriptor provides a Pointer to the root of the red-black tree of memory region objects.

Select one:

- ☐ 1. map_count
- ☒ 2. mm_rb ✓
- ☐ 3. mmap
- ☐ 4. mmap_cache

The correct answer is: mm_rb

Question **154**

Correct

Mark 10.00 out of 10.00

Two signals which cannot be caught by a process are

Select one:

- ☐ 1. SIGSTOP & SIGQUIT
- ☐ 2. SIGQUIT & SIGINT
- ☒ 3. SIGSTOP & SIGKILL ✓
- ☐ 4. SIGKILL & SIGCHLD

The correct answer is: SIGSTOP & SIGKILL

Question **155**

Correct
Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"Page fault exceptions are also caused by programming errors such as access rights violation"

Select one:

- ☒ True ✓
☐ False

The correct answer is 'True'.

Question **156**

Correct
Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"Two adjacent memory regions can be merged even if their access rights do not match."

Select one:

- ☐ True
☒ False ✓

The correct answer is 'False'.

Question **157**

Correct
Mark 5.00 out of 5.00

..... operation associated with superblock object removes every reference to the inode from the VFS data structures and, if the inode no longer appears in any directory, invokes the delete_inode superblock method to delete the inode from the filesystem.

Select one:

- ☐ 1. put_inode(inode)
☐ 2. dirty_inode(inode)
☐ 3. sync_fs(sb, wait)
☒ 4. generic_drop_inode() ✓

The correct answer is: generic_drop_inode()

Question **158**

Correct
Mark 5.00 out of 5.00

..... operation is used as second argument of flock system call to apply write lock on the file.

Select one:

- ☐ 1. LOCK_SH
☒ 2. LOCK_EX ✓
☐ 3. LOCK_UN
☐ 4. SET_LCK

The correct answer is: LOCK_EX

Question **159**

Correct

Mark 5.00 out of 5.00

The technique adopted by Linux to solve the external fragmentation problem is based on the well-known algorithm.

Select one:

- ☐ 1. Dynix allocator
- ☐ 2. Power-of-two free lists
- ☐ 3. McKusick-Karels allocator
- ☒ 4. Buddy system ✓

The correct answer is: Buddy system

Question **160**

Correct

Mark 5.00 out of 5.00

..... can be used inside interrupt handlers and deferrable functions, because requesting such a mapping never blocks the current process and it uses a set of 13 windows or Page Table entries for mapping per CPU.

Select one:

- ☐ 1. Non-Contiguous Memory Allocation
- ☒ 2. Temporary kernel mappings ✓
- ☐ 3. Permanent kernel mappings
- ☐ 4. Contiguous Memory Allocation

The correct answer is: Temporary kernel mappings

Question **161**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"A permanent kernel mapping can be established in interrupt handlers and deferrable functions."

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **162**

Correct

Mark 5.00 out of 5.00

The function releases page frames to the per-CPU page frame caches

Select one:

- ☐ 1. __free_pages_bulk()
- ☐ 2. buffered_rmqueue()
- ☒ 3. free_hot_cold_page() ✓
- ☐ 4. __rmqueue()

The correct answer is: free_hot_cold_page()

Question **163**

Correct

Mark 5.00 out of 5.00

The macro yields the address of the page descriptor associated with the linear address `addr`.

Select one:

- ☐ 1. `page_to_pfn(addr)`
- ☐ 2. `pfn_to_page(pfn)`
- ☐ 3. `page_to_virt(pfn)`
- ☒ 4. `virt_to_page(addr)` ✓

The correct answers are: `virt_to_page(addr)`, `pfn_to_page(pfn)`

Question **164**

Correct

Mark 10.00 out of 10.00

SGL's and IBM's, limit themselves to logging the operations affecting metadata.

Select one:

- ☐ 1. JFS, XFS
- ☐ 2. JFS, VFS
- ☐ 3. VFS, XFS
- ☒ 4. XFS, JFS ✓

The correct answer is: XFS, JFS

Question **165**

Correct

Mark 5.00 out of 5.00

The field of the memory descriptor provides a Pointer to the last referenced memory region object.

Select one:

- ☒ 1. `mmap_cache` ✓
- ☐ 2. `mm_rb`
- ☐ 3. `mmap`
- ☐ 4. `map_count`

The correct answer is: `mmap_cache`

Question **166**

Correct

Mark 5.00 out of 5.00

The function releases noncontiguous memory areas created by `vmalloc()`

Select one:

- ☐ 1. `kmem_cache_alloc()`
- ☒ 2. `vfree()` ✓
- ☐ 3. `vmalloc()`
- ☐ 4. `kmem_cache_free()`

The correct answer is: `vfree()`

Question **167**

Correct

Mark 10.00 out of 10.00

The following command is used to set the setgid and reset the group-execute permission bit of the file myscript.

Select one:

- ☐ 1. chmod 4764 myscript
- ☐ 2. chmod 4777 myscript
- ☐ 3. \$ chmod 2777 myscript
- ☒ 4. \$ chmod 2764 myscript ✓

The correct answer is: \$ chmod 2764 myscript

Question **168**

Correct

Mark 5.00 out of 5.00

The function allocates a noncontiguous memory area to the kernel.

Select one:

- ☐ 1. vfree()
- ☐ 2. kmem_cache_alloc()
- ☐ 3. kmem_cache_free()
- ☒ 4. vmalloc() ✓

The correct answer is: vmalloc()

Question **169**

Correct

Mark 5.00 out of 5.00

Inside each memory zone, page frames are handled by a component named

Select one:

- ☐ 1. Slab Allocator
- ☐ 2. Zone Allocator
- ☐ 3. Per-CPU Page frame cache
- ☒ 4. Buddy System ✓

The correct answer is: Buddy System

Question **170**

Correct

Mark 5.00 out of 5.00

..... is a function definition that specifies how to obtain a given service.

Select one:

- ☒ 1. Application Programmer Interface (API) ✓
- ☐ 2. Re-entrant
- ☐ 3. System call
- ☐ 4. Signal

The correct answer is: Application Programmer Interface (API)

Question **171**

Correct

Mark 5.00 out of 5.00

If of 3-bit error_code parameter of do_page_fault() function is set, the exception was caused by an invalid access right.

Select one:

- ☐ 1. bit 2
- ☒ 2. bit 0 ✓
- ☐ 3. bit 1
- ☐ 4. bit 3

The correct answer is: bit 0

Question **172**

Correct

Mark 5.00 out of 5.00

Categorize the HFS file system into class.

Select one:

- ☒ 1. Disk-based file system ✓
- ☐ 2. Network file system
- ☐ 3. None of these classes
- ☐ 4. Special file system

The correct answer is: Disk-based file system

Question **173**

Correct

Mark 5.00 out of 5.00

Given a main memory of 4GB, determine the maximum number of page frames in this memory considering Linux default page frame size.

Select one:

- ☐ 1. 4 Mega page frames
- ☒ 2. 1Mega page frames ✓
- ☐ 3. 1Giga page frames
- ☐ 4. 1Kilo page frames

The correct answer is: 1Mega page frames

Question **174**

Correct

Mark 5.00 out of 5.00

"Mandatory locks are enforced even between noncooperative processes."

SAY TRUE OR FALSE.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **175**

Correct

Mark 5.00 out of 5.00

..... field in the zone descriptor provides pointer to first page descriptor of the zone.

Select one:

- ☒ 1. zone_mem_map ✓
- ☐ 2. zone_start_pfn
- ☐ 3. zone_pgdat
- ☐ 4. zone_active_list

The correct answer is: zone_mem_map

Question **176**

Correct

Mark 5.00 out of 5.00

..... field in inode object provides pointer to default file operations

Answer: ✓

The correct answer is: i_fop

Question **177**

Correct

Mark 5.00 out of 5.00

The field of the memory descriptor provides number of page frames allocated to the Process.

Select one:

- ☐ 1. hiwater_rss
- ☒ 2. rss ✓
- ☐ 3. total_vm
- ☐ 4. mmlist

The correct answer is: rss

Question **178**

Correct

Mark 5.00 out of 5.00

Kernel uses function to destroy a permanent kernel mapping

Select one:

- ☐ 1. kmap()
- ☒ 2. kunmap() ✓
- ☐ 3. kunmap_atomic
- ☐ 4. kmap_atomic()

The correct answer is: kunmap()

Question **179**

Correct

Mark 5.00 out of 5.00

..... operation associated with file object flushes the file by writing all cached data to disk.

Select one:

- ☐ 1. poll(file, poll_table)
- ☐ 2. mmap(file, vma)
- ☐ 3. ioctl(inode, file, cmd, arg)
- ☒ 4. fsync(file, dentry, flag) ✓

The correct answer is: fsync(file, dentry, flag)

Question **180**

Correct

Mark 5.00 out of 5.00

Categorize the UFS file system into class.

Select one:

- ☐ 1. None of these classes
- ☐ 2. Network file system
- ☐ 3. Special file system
- ☒ 4. Disk-based file system ✓

The correct answer is: Disk-based file system

Question **181**

Correct

Mark 5.00 out of 5.00

The function releases an object previously allocated by the slab allocator to some kernel function.

Select one:

- ☐ 1. vmalloc()
- ☐ 2. kmem_cache_alloc()
- ☒ 3. kmem_cache_free() ✓
- ☐ 4. vfree()

The correct answer is: kmem_cache_free()

Question **182**

Correct

Mark 5.00 out of 5.00

The function insert_vm_struct() in turn invokes the function for Inserting a memory region in the linked list and the red-black tree of the memory descriptor.

Select one:

- ☐ 1. do_munmap()
- ☐ 2. do_mmap()
- ☐ 3. vma_unlink()
- ☒ 4. vma_link() ✓

The correct answer is: vma_link()

Question **183**

Correct

Mark 5.00 out of 5.00

..... field in superblock object provides Name of the block device containing the superblock

Select one:

- ☒ 1. s_id ✓
- ☐ 2. s_op
- ☐ 3. s_type
- ☐ 4. s_fs_info

The correct answer is: s_id

Question **184**

Correct

Mark 5.00 out of 5.00

The field of the memory descriptor provides a Pointer to the head of the list of memory region objects.

Select one:

- ☐ 1. mmap_cache
- ☐ 2. mm_rb
- ☐ 3. map_count
- ☒ 4. mmap ✓

The correct answer is: mmap

Question **185**

Correct

Mark 5.00 out of 5.00

.....is a short message sent to a process by the kernel to notify the process that certain event has occurred.

Select one:

- ☐ 1. Kernel Control path
- ☐ 2. System call
- ☒ 3. Signal ✓
- ☐ 4. API

The correct answer is: Signal

Question **186**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"In a traditional Unix system, there is only one tree of mounted filesystems: starting from the system's root filesystem, each process can potentially access every file in a mounted filesystem by specifying the proper pathname.

In this respect, Linux is more refined: every process might have its own tree of mounted filesystems the so-called namespace of the process."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **187**

Correct

Mark 5.00 out of 5.00

The field is the main usage counter of the memory descriptor where all "users" in mm_users count as one unit.

Select one:

- ☒ 1. mm_count ✓
- ☐ 2. mm_users
- ☐ 3. mm_alloc()
- ☐ 4. kmem_cache_alloc()

The correct answer is: mm_count

Question **188**

Correct

Mark 5.00 out of 5.00

..... operation associated with superblock object is Invoked when flushing the filesystem to update filesystem-specific data structures on disk (used by journaling filesystems).

Select one:

- ☒ 1. sync_fs(sb, wait) ✓
- ☐ 2. generic_drop_inode()
- ☐ 3. dirty_inode(inode)
- ☐ 4. put_inode(inode)

The correct answer is: sync_fs(sb, wait)

Question **189**

Correct

Mark 5.00 out of 5.00

In NUMA 32-bit architecture, the number of bits required in flags field to store both zone and node numbers is

Select one:

- ☐ 1. 10
- ☒ 2. 8 ✓
- ☐ 3. 3
- ☐ 4. 12

The correct answer is: 8

Question **190**

Correct

Mark 5.00 out of 5.00

A kernel function gets dynamic memory by invoking to get pages from the zoned page frame allocator

Select one:

- ☐ 1. kmem_cache_alloc() or kmalloc()
- ☐ 2. vmalloc()
- ☐ 3. vmalloc_32()
- ☒ 4. __get_free_pages() ✓

The correct answer is: __get_free_pages()

Question **191**

Correct

Mark 5.00 out of 5.00

An signal is always delivered, and there is no further action.

Select one:

- ☐ 1. fatal
- ☐ 2. blocked
- ☒ 3. ignored ✓
- ☐ 4. unblocked

The correct answer is: ignored

Question **192**

Correct

Mark 5.00 out of 5.00

The function changes the size of a memory area previously allocated by malloc() or calloc().

Select one:

- ☐ 1. sys_brk(addr)
- ☒ 2. realloc(ptr,size) ✓
- ☐ 3. calloc(n,size)
- ☐ 4. brk()

The correct answer is: realloc(ptr,size)

Question **193**

Correct

Mark 5.00 out of 5.00

Categorize the XFS file system into class.

Select one:

- ☒ 1. Disk-based file system ✓
- ☐ 2. Network file system
- ☐ 3. Special file system
- ☐ 4. None of these classes

The correct answer is: Disk-based file system

Question **194**

Correct

Mark 5.00 out of 5.00

..... field in dentry object provides pointers for the list of unused dentries

Select one:

- ☐ 1. d_inode
- ☐ 2. d_op
- ☐ 3. d_sb
- ☒ 4. d_lru ✓

The correct answer is: d_lru

Question **195**

Correct

Mark 5.00 out of 5.00

All information related to the process address space is included in an object called the memory descriptor of type

Select one:

- ☐ 1. vm_area_struct
- ☒ 2. mm_struct ✓
- ☐ 3. mem_map
- ☐ 4. page_dat_t

The correct answer is: mm_struct

Question **196**

Correct

Mark 5.00 out of 5.00

The field of the memory region object or descriptor provides a Pointer to the memory descriptor that owns that memory region.

Select one:

- ☒ 1. vm_mm ✓
- ☐ 2. vm_start
- ☐ 3. vm_next
- ☐ 4. vm_end

The correct answer is: vm_mm

Question **197**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"If the page has the Read or Execute access right but does not have either the Write or the Share access right, the Read/Write bit is set."

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **198**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"While the system boots, the kernel finds the major number of the disk that contains the root filesystem in the ROOT_DEV variable."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **199**

Correct

Mark 5.00 out of 5.00

..... state of dentry object indicates that the dentry object is not currently used by the kernel. The d_count usage counter of the object is 0, but the d_inode field still points to the associated inode. The dentry object contains valid information, but its contents may be discarded if necessary in order to reclaim memory.

Answer: unused



The correct answer is: Unused

Question **200**

Correct

Mark 5.00 out of 5.00

The is an abstraction layer between the application program and the filesystem implementations

Answer: VFS



The correct answer is: VFS

Question **201**

Correct

Mark 5.00 out of 5.00

..... operation associated with superblock object is Invoked when the inode is released its reference counter is decreased to perform filesystem-specific operations.

Select one:

- ☐ 1. sync_fs(sb, wait)
- ☐ 2. generic_drop_inode()
- ☐ 3. dirty_inode(inode)
- ☒ 4. put_inode(inode) ✓

The correct answer is: put_inode(inode)

Question **202**

Correct

Mark 5.00 out of 5.00

State information of a page frame is kept in a page descriptor of type

Answer: page



The correct answer is: page

Question **203**

Correct

Mark 5.00 out of 5.00

The slab allocator takes advantage of the free unused bytes to the slab which is used simply to subdivide the slabs and allow the memory allocator to spread objects out among different linear addresses. In this way, the kernel obtains the best possible performance from the microprocessor's hardware cache.

Answer: color



The correct answer is: color

Question **204**

Correct

Mark 5.00 out of 5.00

When a process deletes a file or truncates it to 0 length, all its data blocks must be reclaimed. This is done by, which receives the address of the file's inode object as its parameter.

Select one:

- ☐ 1. ext2_get_block()
- ☐ 2. ext2_alloc_block()
- ☐ 3. ext2_free_blocks()
- ☒ 4. ext2_truncate() ✓

The correct answer is: ext2_truncate()

Question **205**

Correct

Mark 5.00 out of 5.00

Given a main memory of 8GB, determine the maximum number of page frames in this memory considering Linux default page frame size.

Select one:

- ☐ 1. 4 Kilo page frames
- ☐ 2. 2 Kilo page frames
- ☐ 3. 2 Giga page frames
- ☒ 4. 2Mega page frames ✓

The correct answer is: 2Mega page frames

Question **206**

Correct

Mark 5.00 out of 5.00

Linux's kernel memory allocator uses Slab Allocator algorithm on top of a approach

Select one:

- ☐ 1. Dynix allocator
- ☐ 2. Power-of-two free lists
- ☐ 3. McKusick-Karels allocator
- ☒ 4. Buddy system ✓

The correct answer is: Buddy system

Question **207**

Correct

Mark 5.00 out of 5.00

..... operation associated with inode object searches a directory for an inode corresponding to the filename included in a dentry object.

Select one:

- ☒ 1. lookup(dir, dentry, nameidata) ✓
- ☐ 2. link(old_dentry, dir, new_dentry)
- ☐ 3. follow_link(inode, nameidata)
- ☐ 4. mknod(dir, dentry, mode, rdev)

The correct answer is: lookup(dir, dentry, nameidata)

Question **208**

Correct

Mark 5.00 out of 5.00

In Linux, the function used for finding the closest region to a given address is

Select one:

- ☒ 1. find_vma() ✓
- ☐ 2. get_unmapped_area()
- ☐ 3. find_vma_intersection()
- ☐ 4. insert_vm_struct()

The correct answer is: find_vma()

Question **209**

Correct

Mark 5.00 out of 5.00

..... operation associated with file object performs a memory mapping of the file into a process address space

Select one:

- ☐ 1. poll(file, poll_table)
- ☐ 2. ioctl(inode, file, cmd, arg)
- ☐ 3. fsync(file, dentry, flag)
- ☒ 4. mmap(file, vma) ✓

The correct answer is: mmap(file, vma)

Question **210**

Correct

Mark 5.00 out of 5.00

Linux implements a memory region by means of an object of type

Select one:

- ☒ 1. vm_area_struct ✓
- ☐ 2. mem_map
- ☐ 3. mm_struct
- ☐ 4. page_dat_t

The correct answer is: vm_area_struct

Question **211**

Correct

Mark 5.00 out of 5.00

In Linux, the function used for Allocating a Linear Address Interval is

Select one:

- ☒ 1. do_mmap() ✓
- ☐ 2. vma_unlink()
- ☐ 3. do_munmap()
- ☐ 4. vma_link()

The correct answer is: do_mmap()

Question **212**

Correct

Mark 5.00 out of 5.00

The size of a NORMAL zone is

Select one:

- ☒ 1. 896MB ✓
- ☐ 2. 128MB
- ☐ 3. 16MB
- ☐ 4. 256MB

The correct answer is: 896MB

Question **213**

Correct

Mark 5.00 out of 5.00

..... in each memory zone stores page frames whose contents are likely to be included in the CPU's hardware cache.

Select one:

- ☒ 1. A hot cache ✓
- ☐ 2. A cold cache
- ☐ 3. Split Cache
- ☐ 4. Unified Cache

The correct answer is: A hot cache

Question **214**

Correct

Mark 5.00 out of 5.00

..... field in the page descriptor encodes the node and zone number to which the page frame belongs.

Answer:



The correct answer is: flags

Question **215**

Correct

Mark 5.00 out of 5.00

In 80x86 architecture, in the fourth gigabyte of RAM, PAGE_OFFSET refers to address

Select one:

- ☐ 1. 0xFE000000
- ☐ 2. 0xFC000000
- ☐ 3. 0x10000000
- ☒ 4. 0xC0000000 ✓

The correct answer is: 0xC0000000

Question **216**

Correct

Mark 5.00 out of 5.00

The head of the red-black tree is referenced by the field of the memory descriptor.

Answer:



The correct answer is: mm_rb

Question **217**

Correct

Mark 5.00 out of 5.00

The field of the memory region object or descriptor provides first linear address inside the memory region.

Select one:

- ☐ 1. vm_next
- ☐ 2. vm_end
- ☒ 3. vm_start ✓
- ☐ 4. vm_mm

The correct answer is: vm_start

Question **218**

Correct

Mark 10.00 out of 10.00

Each node has a descriptor of type and all node descriptors are stored in a singly linked list, whose first element is pointed to by the variable.

Select one:

- ☐ 1. node_mem_map, node_start_pfn
- ☐ 2. node_mem_map, pgdat_list
- ☐ 3. pg_data_t, node_start_pfn
- ☒ 4. pg_data_t, pgdat_list ✓

The correct answer is: pg_data_t, pgdat_list

Question **219**

Correct

Mark 5.00 out of 5.00

In NUMA 64-bit architecture, the number of bits required in flags field to store both zone and node numbers is

Select one:

- ☐ 1. 3
- ☐ 2. 10
- ☒ 3. 12 ✓
- ☐ 4. 8

The correct answer is: 12

Question **220**

Correct

Mark 5.00 out of 5.00

The function is invoked to get a new memory descriptor.

Select one:

- ☐ 1. mm_users
- ☐ 2. mm_count
- ☐ 3. kmem_cache_alloc()
- ☒ 4. mm_alloc() ✓

The correct answer is: mm_alloc()

Question **221**

Correct

Mark 5.00 out of 5.00

In 32-bit architecture, allow the kernel to establish long-lasting mappings of high-memory page frames into the kernel address space using a dedicated Page Table in the master kernel page tables. But this mapping may block the current process when no free Page Table entries exist that can be used as "windows" on the page frames in high memory.

Select one:

- ☐ 1. Temporary kernel mappings
- ☐ 2. Contiguous Memory Allocation
- ☒ 3. Permanent kernel mappings ✓
- ☐ 4. Non-Contiguous Memory Allocation

The correct answer is: Permanent kernel mappings

Question **222**

Correct

Mark 5.00 out of 5.00

Categorize the ReiserFS file system into class.

Select one:

- ☒ 1. Disk-based file system ✓
- ☐ 2. Network file system
- ☐ 3. Special file system
- ☐ 4. None of these classes

The correct answer is: Disk-based file system

Question **223**

Correct

Mark 5.00 out of 5.00

The purpose of the function is to split a memory region that intersects a linear address interval into two smaller regions, one outside of the interval and the other inside.

Select one:

- ☐ 1. vma_unlink()
- ☐ 2. vma_link()
- ☐ 3. do_mmap()
- ☒ 4. split_vma() ✓

The correct answer is: split_vma()

Question **224**

Correct

Mark 10.00 out of 10.00

On NUMA 32-bit architectures, flags field in the page descriptor has bits for the zone index and bits for the node number.

Select one:

- ☐ 1. 2, 1
- ☒ 2. 2, 6 ✓
- ☐ 3. 2, 10
- ☐ 4. 2,4

The correct answer is: 2, 6

Question **225**

Correct

Mark 5.00 out of 5.00

To speed up searching, insertion and deletion operations, Linux stores memory region descriptors in data structures called

Select one:

- ☐ 1. hash table
- ☐ 2. doubly linked list
- ☐ 3. singly linked list
- ☒ 4. red-black trees ✓

The correct answer is: red-black trees

Question **226**

Correct

Mark 5.00 out of 5.00

The kernel represents intervals of linear addresses by means of resources called, which are characterized by an initial linear address, a length, and some access rights.

Answer: ✓

The correct answer is: memory regions

Question **227**

Correct

Mark 5.00 out of 5.00

Given a memory area MA2 with start address 0xD0000000 and size 512 MB, compute the end address

Select one:

- ☐ 1. 0xFFFFFFFF
- ☐ 2. 0xCFFFFFFF
- ☒ 3. 0xEFFFFFFF ✓
- ☐ 4. 0xBFFFFFFF

The correct answer is: 0xEFFFFFFF

Question **228**

Correct

Mark 5.00 out of 5.00

..... system call destroys a memory mapping for a file, thus contracting the process address space.

Select one:

- ☐ 1. mmap_cache
- ☐ 2. brk()
- ☐ 3. mmap()
- ☒ 4. munmap() ✓

The correct answer is: munmap()

Question **229**

Correct

Mark 5.00 out of 5.00

Size of each page descriptor is bytes

Select one:

- ☒ 1. 32 ✓
- ☐ 2. 8
- ☐ 3. 64
- ☐ 4. 16

The correct answer is: 32

Question **230**

Correct

Mark 5.00 out of 5.00

All memory descriptors are stored in a

Select one:

- ☐ 1. singly linked list
- ☐ 2. Binary tree
- ☒ 3. doubly linked list ✓
- ☐ 4. hash table

The correct answer is: doubly linked list

Question **231**

Correct

Mark 5.00 out of 5.00

..... operation associated with superblock object is Invoked when the inode is marked as modified and used by filesystems such as ReiserFS and Ext3/4 to update the filesystem journal on disk.

Select one:

- ☐ 1. generic_drop_inode()
- ☐ 2. put_inode(inode)
- ☐ 3. sync_fs(sb, wait)
- ☒ 4. dirty_inode(inode) ✓

The correct answer is: dirty_inode(inode)

Question **232**

Correct

Mark 5.00 out of 5.00

The key idea behind the VFS consists of introducing a model capable of representing all supported filesystems.

Answer: ✓

The correct answer is: common file

Question **233**

Correct

Mark 5.00 out of 5.00

Categorize the UDF file system into class.

Select one:

- ☒ 1. Disk-based file system ✓
- ☐ 2. Network file system
- ☐ 3. Special file system
- ☐ 4. None of these classes

The correct answer is: Disk-based file system

Question **234**

Correct

Mark 10.00 out of 10.00

On NUMA 64-bit architectures, flags field in the page descriptor has bits for the zone index and bits for the node number.

Select one:

- ☐ 1. 2,4
- ☐ 2. 2, 6
- ☒ 3. 2, 10 ✓
- ☐ 4. 2, 1

The correct answer is: 2, 10

Question **235**

Correct

Mark 5.00 out of 5.00

The component of Zoned Page frame allocator named receives the requests for allocation and deallocation of dynamic memory. In the case of allocation requests, the component searches a memory zone that includes a group of contiguous page frames that can satisfy the request.

Select one:

- ☐ 1. Slab Allocator
- ☐ 2. Per-CPU Page frame cache
- ☒ 3. Zone Allocator ✓
- ☐ 4. Buddy System

The correct answer is: Zone Allocator

Question **236**

Correct

Mark 5.00 out of 5.00

..... field in file object provides Pointers for generic file object list.

Select one:

- ☐ 1. f_dentry
- ☐ 2. f_op
- ☒ 3. f_list ✓
- ☐ 4. f_vfsmnt

The correct answer is: f_list

Question **237**

Correct

Mark 5.00 out of 5.00

Ctrl & C key combinations generate signal

Select one:

- ☐ 1. SIGSTP
- ☒ 2. SIGINT ✓
- ☐ 3. SIGKILL
- ☐ 4. SIGQUIT

The correct answer is: SIGINT